

CHM 120

Laboratory Techniques and Measurements

Final Report

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Lesson Laboratory Techniques and Measurements
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Test Your Knowledge

Rank the measurements in order from smallest to largest.

$3.0 \times 10^{-7} \text{ m}$
$2.0 \times 10^{-6} \text{ m}$
$1.0 \times 10^{-5} \text{ m}$
$4.0 \times 10^3 \text{ mm}$
1.0 m

Match each SI base unit to the appropriate measurement.

Length	Mass	Time	Temperature	Amount of Matter
1 meter	2 kilogram	3 second	4 kelvin	5 mol

Perform the following conversions.

345.7 K	Convert 61° F to °C	1	16° C
12° C	Convert 19.4° C to K	2	292.6 K
2.53 cm	Convert 15.3 mm to cm	3	1.53 cm
0.425 m	Convert 4.25 cm to m	4	0.0425 m
2.184 kg	Convert 218.4 g to kg	5	0.2184 kg

Exploration

A micrometer is equal to 0.000001 m.

- ☒ True
- ☐ False

The SI system is derived from the ____.

- ☐ United States system
- ☒ metric system
- ☐ English unit system
- ☐ math system

Every measuring device measures length to the same degree of accuracy and precision.

- ☐ True
- ☒ False

The SI system unit for temperature is ____.

- ☐ degrees Celsius
- ☐ degrees Fahrenheit
- ☒ kelvin

One liter is equal to ____.

- ☐ 1 cm³
- ☐ 10 cm³
- ☐ 100 cm³
- ☒ 1,000 cm³

A meniscus is the curve that forms at the surface of a liquid.

- ☒ True
- ☐ False

Graduated cylinders and graduated pipets are designed to measure ____ volume(s).

- ☒ a range of
- ☐ only one

Archimedes' method is used to measure the volume of a(n) ____.

- ☐ square
- ☐ rectangle
- ☒ irregularly shaped object

Density is derived from SI units of _____ and _____.

- ☐ volume; length
- ☐ mass; amount of substance
- ☐ length; mass
- ☒ mass; volume

In the equation $C_1 \times V_1 = C_2 \times V_2$, C_1 represents the ____ of the solution.

- ☒ initial concentration
- ☐ initial volume
- ☐ final concentration
- ☐ final volume

The final digit in any measurement is the estimated, uncertain digit.

- ☒ True
- ☐ False

Exercise 1

Water boils at 100°C at sea level. If the water in this experiment did not boil at 100°C, what could be the reason?

B /

18 Word(s)

Because I used tap water, there are naturally more things in it that could disturb the temperature overall.

While heating two different samples of water at sea level, one boils at 102°C and one boils at 99.2°C. Calculate the percent error for each sample from the theoretical 100.0°C.

B
/
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12 Word(s)

First sample, the error is 2%.
Second sample, the error is 0.8%

Data Table 1: Length Measurements

	Length (cm)	Length (mm)	Length (m)
CD or DVD	12	120	.12
Key	6	60	.06
Spoon	20	200	.20
Fork	20	200	.20

Data Table 2: Temperature Measurements

	Temperature (°C)	Temperature (°F)	Temperature (K)
Hot from Tap	44.1	111.4	384.55
Boiling	100.5	212.9	486.05
Boiling for 5 minutes	102.5	216.5	489.65
Cold from Tap	19.8	67.64	340.79
Ice Water - 1 minute	45.3	113.54	386.69
Ice Water - 5 minutes	9.0	48.2	321.35

Data Table 3: Mass Measurements

	Estimated Mass (g)	Actual Mass (g)	Actual Mass (kg)
Pen or Pencil	8	7.28	.00728
3 Pennies	3	8.12	.00812
1 Quarter	5	5.69	.00569
2 Quarters, 3 Dimes	15	18.26	.01826
4 Dimes, 5 Pennies	10	22.76	.02276
3 Quarters, 1 Dime, 5 Pennies	21	33.11	.03311
Key	9	15.27	.01527
Key, 1 Quarter, 4 Pennies	14	32.17	.03217

Exercise 2

An unknown, rectangular substance measures 3.60 cm high, 4.21 cm long, and 1.17 cm wide. If the mass is 21.3 g, what is this substance's density (in grams per milliliter)?

B / <u>U</u>	6 Word(s)
Volume = 17.733 cm^3 Density = 1.201 g/mL	

A sample of gold (Au) has a mass of 26.15 g. Given that the theoretical density is 19.30 g/mL, what is the volume of the gold sample?

B / <u>U</u>	1 Word(s)
$1.35 \text{ cm}^{310.4}$	

A student was given an unknown metal. The student determined that the mass of the metal was 30.2 g. The student placed the metal in a graduated cylinder filled with 20.0 mL of water. The metal increased the volume of water to 22.9 mL. Calculate the density of the metal and determine the identity of the metal using the table below.

Metal	Density (g/mL)
Lead	11.3
Silver	10.5
Nickel	9.90
Zinc	7.14

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10.41g/mL and it is silver

Data Table 4: Liquid Measurements

	Water	Isopropyl Alcohol
Mass A: Graduated Cylinder (g)	17.20	17.20
Volume (mL)	5.0	5.0
Mass B: Graduated Cylinder with Liquid (g)	22.01	21.115
Mass B-A: Liquid (g)	4.81	3.915
Density (g/mL)	.962	.783
Percent Error (%)	3.8	.38

Data Table 5: Magnet - Direct Measurement Method

	Magnet
Mass (g)	4.34
Length (cm)	2.5
Width (cm)	.5
Height (cm)	.5
Volume (cm ³)	.625
Density (g/cm ³)	6.94

Data Table 6: Water Displacement Method

	Magnet	Metal Bolt
Mass (g)	4.24	7.54
Initial Volume of Graduated Cylinder (mL)	7.0	7.0
Final Volume of Graduated Cylinder (mL)	8.0	8.5
Object Volume (mL)	1.0	1.5
Density (g/mL)	4.24	5.027

Exercise 3

How would you prepare 10.0 mL of a 0.25% *m/v* HCl solution if 1% *m/v* HCl was available? How much 1% *m/v* HCl is needed? How much distilled water is used?

B / U
0 Word(s)

From the graph of Density vs. Concentration, created in Graph 1, what was the relationship between the concentration of the sugar solution and the density of the sugar solution?

B / U
27 Word(s)

Honestly, it is hard to develop a relationship here. I may have did something wrong with my Solution 0 Final Concentration...It looks exponential though to some degree.

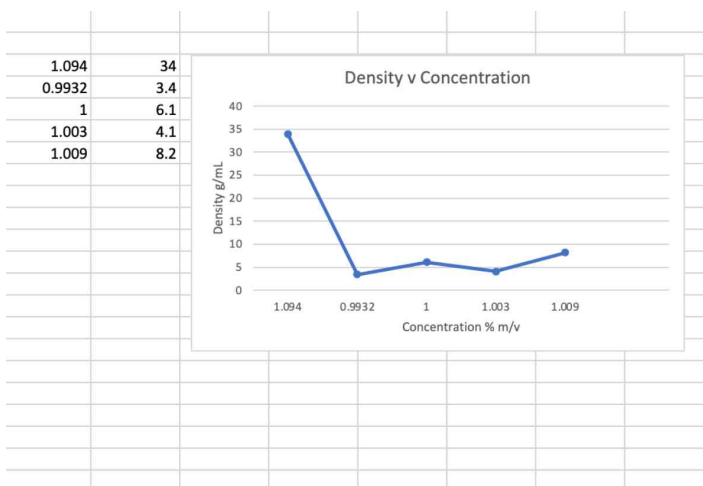
Data Table 8: Initial Concentration

	Sugar (C ₁₂ H ₂₂ O ₁₁)
Mass of Volumetric Flask (g)	24.92
Mass of Sugar (g)	8.05
Total Volume (mL)	25.0
Concentration (% m/V)	34

Data Table 9: Solutions

Solution	Volume (mL)	Mass (g)	Density (g/mL)	Initial Concentration (% m/v)	Volume Transferred (mL)	Final Concentration (% m/v)
0	25.00 mL	27.36	1.094		2.5	34
1	25.00 mL	24.83	.9932	34	2.5	3.4
2	25.00 mL	25.01	1.000	3.4	4.5	6.1
3	25.00 mL	25.07	1.003	6.1	3.0	4.1
4	25.00 mL	25.22	1.009	4.1	6.0	8.2

Graph 1: Density versus Concentration



Competency Review

The SI system unit for the amount of a substance is ____.

- ☒ grams per liter
- ☐ moles
- ☐ Avogadro's number
- ☐ kilograms

The International System of Units (SI) is ____.

- ☐ a scale of measurement
- ☐ the same as U.S. Customary Units
- ☐ the same as English units
- ☒ a standard system to measure length, temperature, time, amount of substance, and mass

Convert 15.00°F to °C.

- ☐ 1.58°C
- ☐ 8.33°C
- ☒ -9.40°C
- ☐ 59°C
- ☐ -8.7°C

Convert 5.00 miles to kilometers.

- ☒ 8.05 km
- ☐ 3.11 km
- ☐ 2.49 km
- ☐ 1.53 km

A meniscus is the curve that forms at the surface of a liquid.

- ☒ True
- ☐ False

_____ is defined as mass per unit of measure.

- ☐ Volume
- ☐ A kilogram
- ☐ Molarity
- ☒ Density
- ☐ Concentration

The definition of % m/V is _____.

- ☐ grams solute/mL solution x 100
- ☒ grams solution/mL solute x 100
- ☐ mL solute/grams solution x 100
- ☐ mL solution/grams solute x 100

Significant figures include only the certain digits of a measurement.

- ☐ True
- ☒ False

When reading a graduated cylinder made of glass, one must read the volume at eye level from the middle of the meniscus.

- ☒ True
- ☐ False

A volumetric flask contains 25.0 mL of a 14% m/V sugar solution. If 2.5 mL of this solution is added to 22.5 mL of distilled water, what is the % m/V of the new solution.

- ☒ 1.4 % m/V
- ☐ 14 % m/V
- ☐ 7.1 % m/V
- ☐ 71 % m/V

Calculate experimental error using the following data: the measured value equals 1.4 cm; the accepted value equals 1.2 cm.

- ☐ -14.3%
- ☒ 14.3%
- ☐ -16.7%
- ☐ 16.7%

What is the volume of an irregularly shaped object that has a mass 3.0 grams and a density of 6.0 g/mL?

- ☐ 1.0 mL
- ☐ 0.5 mL
- ☒ 2.0 mL
- ☐ 18.0 mL

Extension Questions

In the movie, "*Raiders of the Lost Ark*", Indiana Jones takes a gold idol from a cave. The statue is resting on a table which is rigged with a weight sensor. The weight sensor can detect when the weight is removed and will set off a series of unfortunate accidents. To prevent this from happening, Indiana replaces the gold idol with a bag of sand. The volume of the gold idol is approximately 1.0 L. The density of gold is 19.3 g/mL and the density of sand is 2.3 g/mL.

a. Assuming the idol is pure gold, what volume would the bag of sand have to be in order to weigh exactly the same as the idol and not set off the booby-traps?

b. Let's assume that Indiana is successful in removing the idol and returning with it to his laboratory. He decides to determine if it is really pure gold. He weighs the idol and measures the volume by a water displacement method. The results are: mass = 16.5 kg and volume of water displaced = 954 mL. Is the idol made of pure gold? Explain your answer based on the experimental results.

Volume of Gold = .12mL

I do not think it is made of pure gold. It's hard to explain it based on the given numbers. I feel like I am missing something here..